

Tai Shishiba

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EDUCATION

Mount Royal University

Calgary, AB

Bachelor of Science in Computer Science

Sept. 2023 – Expected Apr. 2028

- **Cumulative GPA:** 3.91/4.00
- **Dean's Honour Roll** (6 consecutive semesters)
- **President's Honour Roll** (3 consecutive years)
- **Relevant Courses:** Algorithms and Complexity, System Design, Introduction to Databases, Software Engineering, Web Development, Operating Systems, Human-Computer Interaction, Artificial Intelligence

EXPERIENCE

Freelance Software Developer

Jan. 2026 – Current

All Through The House | Next.js, Prisma, PostgreSQL, Square SDK, Vitest

Okotoks, AB

- Built and maintain a production **B2B consignment-reporting platform**, replacing manual admin reporting with self-serve vendor access across **30+ API routes** and **14 data models**
- **Cut 120+/week vendor sales-data and statement requests** to the business owner, by designing an incremental statement-generation pipeline that converts Square transactions into per-vendor payout statements
- Attributed **7,000+ Square orders** into **15,000+ vendor-specific entries**, by integrating the Square SDK and modeling a vendor concept absent from Square's native API
- Cut analytics API latency **34% (532ms to 347ms)**, by batching five aggregation queries into a single SQL query with materialized CTEs
- Built a **300-case Vitest suite** across 28 modules with zero DB or network dependencies, wiring it into a GitHub Action for fast, deterministic CI runs

Undergraduate Research Assistant

May 2025 – Aug. 2025

Mount Royal University | Python, Numpy, Matplotlib, LaTeX

Calgary, AB

- Built Python visualization pipelines with NumPy and Matplotlib to analyze simulation outputs for malware-propagation research in wireless sensor networks
- Prototyped simulations using emerging research libraries to model malware spread patterns, enabling the team to evaluate new modeling approaches
- Authored LaTeX technical documentation of experimental methodology and results for research publication

PROJECTS

Urban Pulse (HTC 2025) | Next.js, Prisma, PostgreSQL, Ollama, AWS S3 | Github

Oct. 2025 – Oct. 2025

- Built the backend for Urban Pulse, an urban incident-reporting system, in a 4-person team during a hackathon focused on software for positive social impact
- Enabled secure report CRUD and pagination by building **a dozen** type-safe Next.js server actions with per-user authorization and server-side validation
- Designed a 1-to-5 star rating system with composite-key vote upserts, denormalizing each report's average rating for fast reads
- Integrated a vision LLM (**Llama 3.2 11B**) via an OpenAI-compatible API, letting users report issues with just a photo – auto-generating descriptions with prompt engineering and moderation fallbacks

TECHNICAL SKILLS

Languages: JavaScript/Typescript, Python, SQL, Java, C/C++, HTML/CSS, Assembly, Nix

Developer Tools: Git, GitHub, VS Code, Postman, Emacs, Claude, Sentry

Testing: Playwright, Vitest, Pytest, GitHub Actions

Frameworks/Libraries: React, Express, NextJS, Convex, Neon, Prisma, Drizzle, Hono, WorkOS, Clerk, TailwindCSS, NumPy, Matplotlib, Shadcn, Radix UI, BetterAuth, Square SDK, Resend, Axiom